User Manual

FLIM IMAGER



User Manual – Version v2.1.2
12 Nov 2026

Contents

1 General Information **2**
1.1 Introduction . . . 2
1.2 Scope of the Manual . . . 2
1.2.1 Version Note . . . 2
1.3 User Qualification . . . 2
1.4 Developer Information . . . 2

2 Getting Started **3**
2.1 System Requirements . . . 3
2.2 Software Installation . . . 3
2.3 Hardware Connection . . . 3
2.4 Initial Configuration . . . 3
2.4.1 Configuration Wizard . . . 3
2.5 GUI Overview . . . 9
2.6 Scouting & FLIM Modes . . . 10

3 Software Operation **12**
3.1 Startup Procedure . . . 12
3.2 Data Display . . . 12
3.2.1 Channel Configuration . . . 12
3.2.2 Plot Interaction Features . . . 13
3.2.3 Color Maps . . . 13
3.2.4 Intensity Tracing . . . 14
3.3 Data Acquisition . . . 14
3.3.1 Acquisition Parameters . . . 15
3.4 Data Export . . . 15
3.5 Configuration Options . . . 16
3.5.1 Configuration Selection . . . 16
3.5.2 Hardware Setup . . . 16
3.5.3 Laser Synchronization . . . 17
3.5.4 Scan Area . . . 17
3.6 Image Reconstruction . . . 20

4 FLIM Analysis Modes **22**
4.1 InstaFLIM . . . 22
4.2 TCSPC . . . 22
4.3 Calibration . . . 24
4.4 Phasors . . . 24

5 Advanced Operation **27**
5.1 Console Messages . . . 27
5.2 Data Processing Scripts . . . 27

6 Troubleshooting **29**
6.1 Common Issues . . . 29

7 Safety, License and Support **30**
7.1 Safety and Operational Limitations . . . 30
7.2 License . . . 30
7.3 Contact and Support . . . 30

8 Technical Reference **31**
8.1 File Format . . . 31

1 General Information

1.1 Introduction

FLIM IMAGER is a software platform for real-time image acquisition, reconstruction, visualization, and analysis in fluorescence lifetime imaging applications. It provides an intuitive interface for exploring fluorescence lifetime data and supports tools such as customizable colormaps, contrast adjustment, and region-based analysis.

The software supports Time-Correlated Single Photon Counting (TCSPC) and phasor analysis and is designed to operate with the FLIM Data Acquisition Card. It offers multiple acquisition modes, real-time and post-processing analysis, histogram functions, ROI analysis, phasor clusterization, flexible reconstruction modes, and export of data for further analysis using Python or MATLAB scripts.

1.2 Scope of the Manual

This manual covers installation, connection, initial configuration, software operation, parameter configuration, data export, and error handling to support correct use of the FLIM IMAGER software with compatible FLIM LABS hardware. For further information, refer to the manual of the FLIM Data Acquisition Card.

1.2.1 Version Note

This manual describes the software **FLIM IMAGER version 2.0** with the document version v2.1.2 from 12 Nov 2026.

Other software versions may differ in functionality and appearance. If this document does not correspond to your installed software, refer to the official website at https://www.flimlabs.com/products/FLIM-Imager/ to obtain the appropriate version of the software or the corresponding manual.

1.3 User Qualification

This software is intended for qualified laboratory personnel and researchers familiar with data acquisition and imaging hardware and software. It assumes a basic understanding of photon counting and acquisition parameters. For introductory material on these topics, refer to the learning resources available on our website (www.flimlabs.com).

1.4 Developer Information

FLIM LABS makes the source code of FLIM IMAGER publicly available on GitHub to support transparency and collaboration with the research community:

URL NEEDED https://github.com/flim-labs/

You can explore the code, report issues, suggest features, and propose improvements through the repository. This collaborative approach helps adapt the software to specific research needs. For customized acquisition workflows, analysis tools, or integration with your experimental setup, contact FLIM LABS to discuss your requirements and available development support.

2 Getting Started

This chapter describes the prerequisites and initial steps required to install the software, connect the hardware, and perform the first configuration.

2.1 System Requirements

The following specifications are required for running FLIM IMAGER on a PC:

- PC with multicore CPU
- 8 GB RAM
- USB interface for connecting the DAQ to the PC
- Windows 10 or later

Additionally, Python version 3.9 or later is required for advanced features such as analysis scripts. For further details, see Chapter 5.

2.2 Software Installation

FLIM IMAGER can be downloaded from the website: https://www.flimlabs.com/products/FLIM-imager.
If multiple versions are available, select the version that provides the required functionality. To install the software, proceed as follows:

1. Download the selected version. The file is provided as a Windows installer package.
2. Double-click the installer and follow the on-screen instructions.

The Windows installer already includes the ZestSC3 drivers required for the correct operation of the FLIM Data Acquisition Card.

2.3 Hardware Connection

Using this software for data acquisition requires the FLIM Data Acquisition Card. Connect the device as follows:

- Connect the FLIM LABS card to the acquisition PC.
- Verify that the card is correctly recognized by the operating system.
- Connect input, trigger, and reference signals as required for your application.

Figure 2.1 shows an example of a setup including all signal connections required for fluorescence lifetime analysis and full image reconstruction.
For detailed instructions on hardware connections, refer to the user manual for the data acquisition card available on the website: https://www.flimlabs.com/products/FLIM-DAQ.

2.4 Initial Configuration

After completing the installation and all hardware connections, start the software. Figure 2.2 shows the screen displayed upon software startup. Selecting *Guided Configuration* starts a step-by-step process that guides the user through the initial configuration. This process prepares the software for correct operation with the connected hardware and measurement setup.
Alternatively, the software can be started without a configuration by selecting *Blank Project* or a previous configuration can be loaded from a folder.
Recently generated and saved configurations can be loaded by clicking one of the panels shown below *Recent Configurations:*.

2.4.1 Configuration Wizard

The *Guided Configuration Wizard* helps the user to select all parameters needed for configuration of the software for data acquisition and image reconstruction. Simply follow the instructions provided at each step and confirm by clicking on Next to move to the next step.

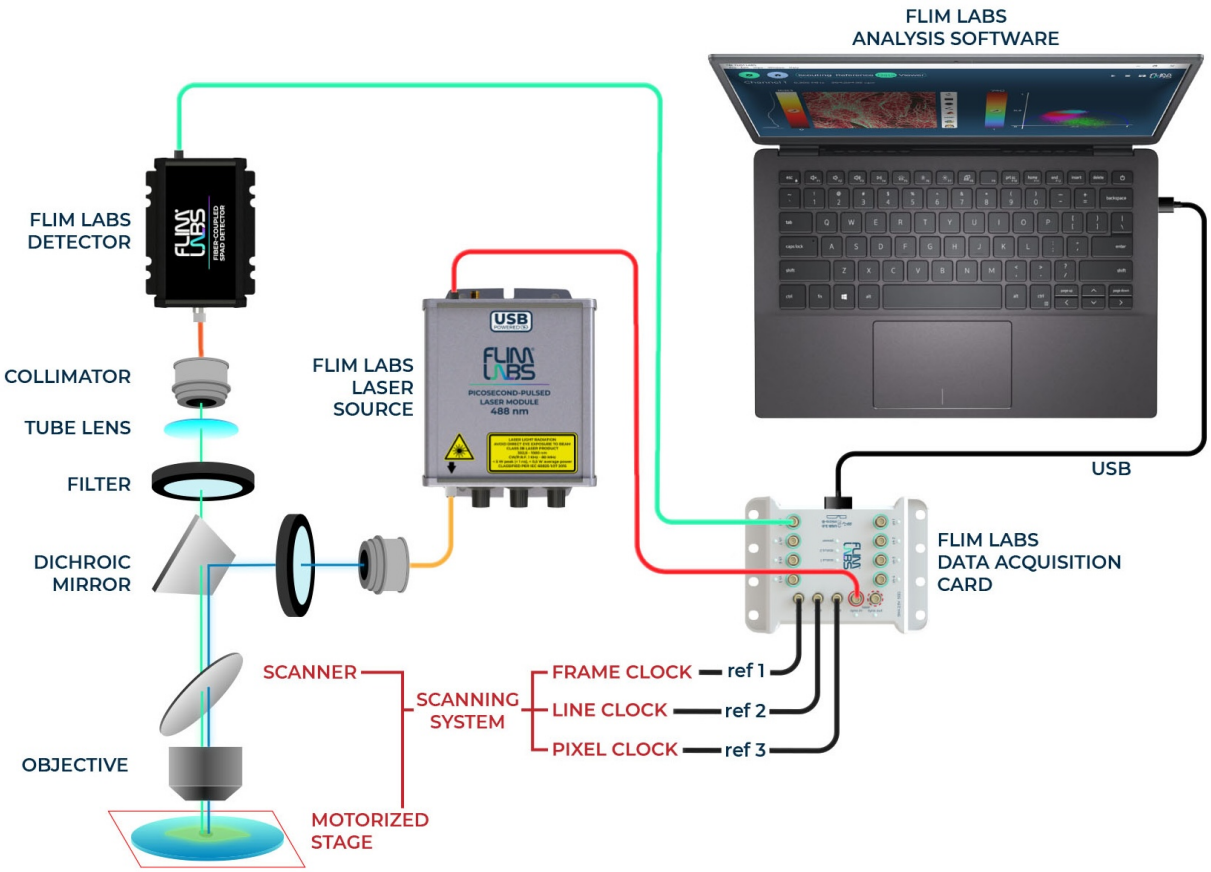


Figure 2.1: Example of a setup including all signal connections required for fluorescence lifetime analysis and full image reconstruction.

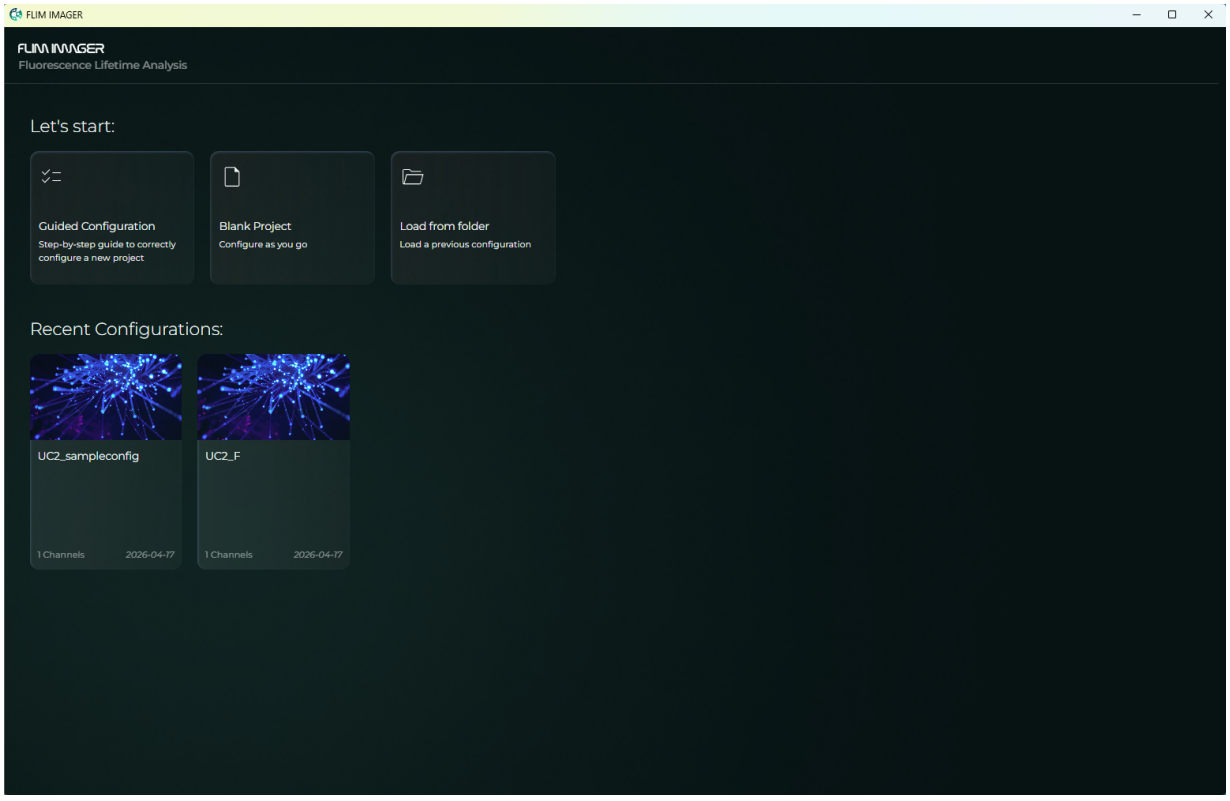
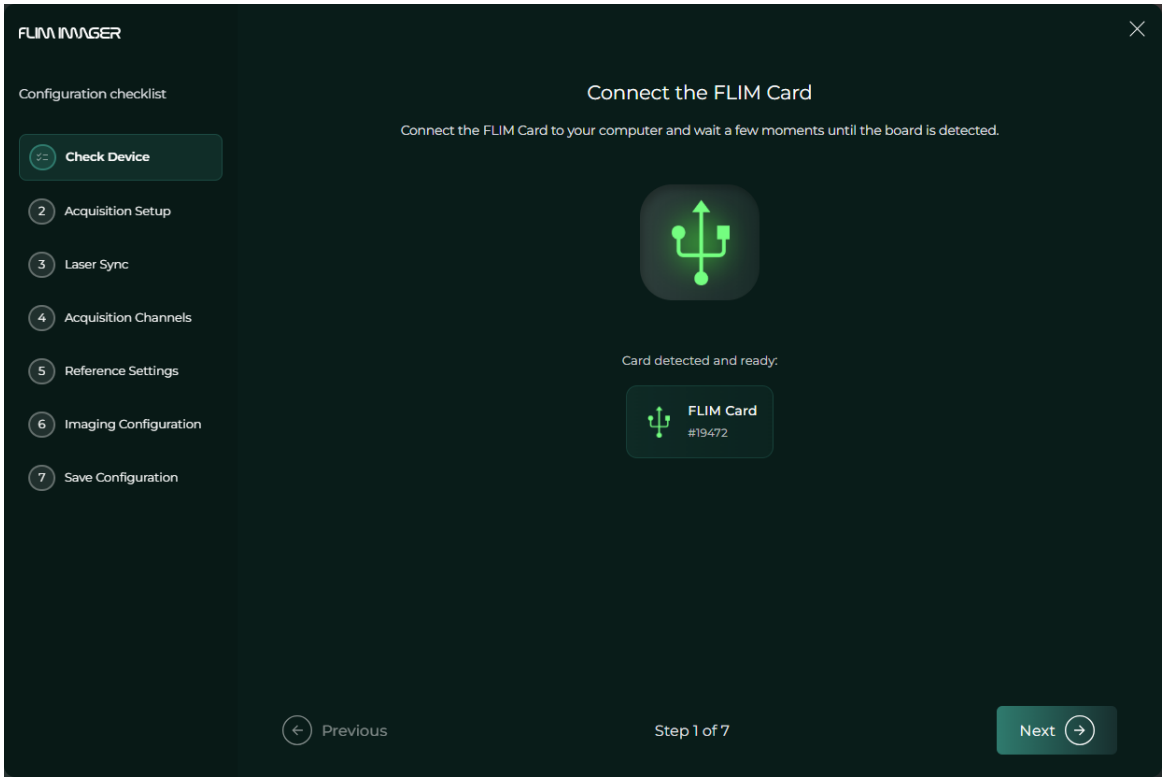


Figure 2.2: Screen shown upon startup of FLIM IMAGER.

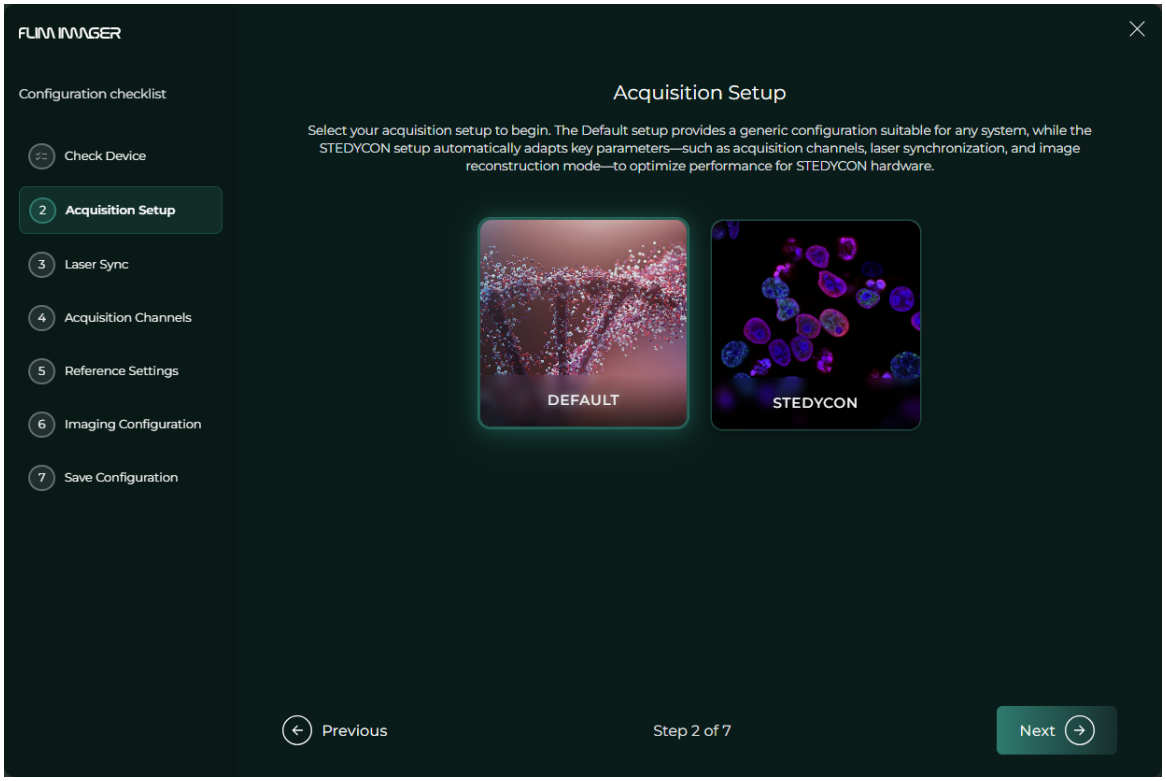
1. Connect the FLIM Data Acquisition Card

Please connect the *FLIM Data Acquisition Card* to the PC. The software will automatically scan for available devices.



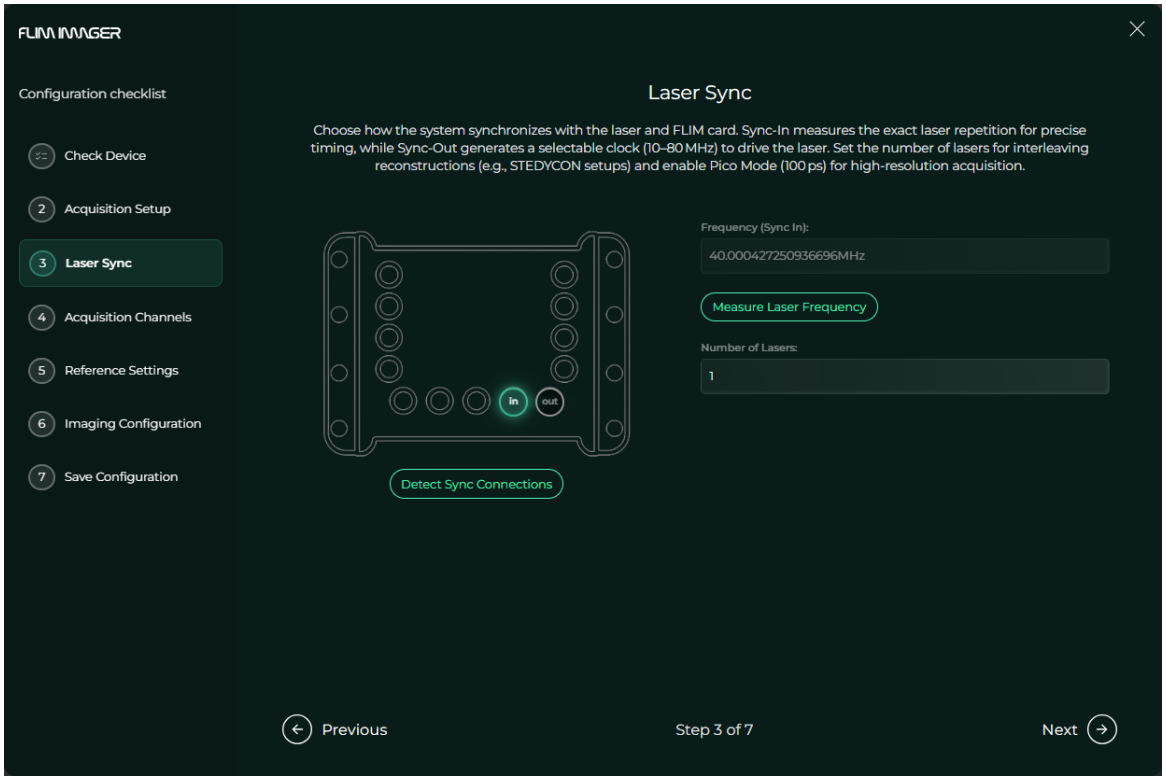
2. Choose default setup mode

Choose a set of pre-configured parameters if your setup matches one of the configurations shown on the screen.



3. Laser Synchronization Setting

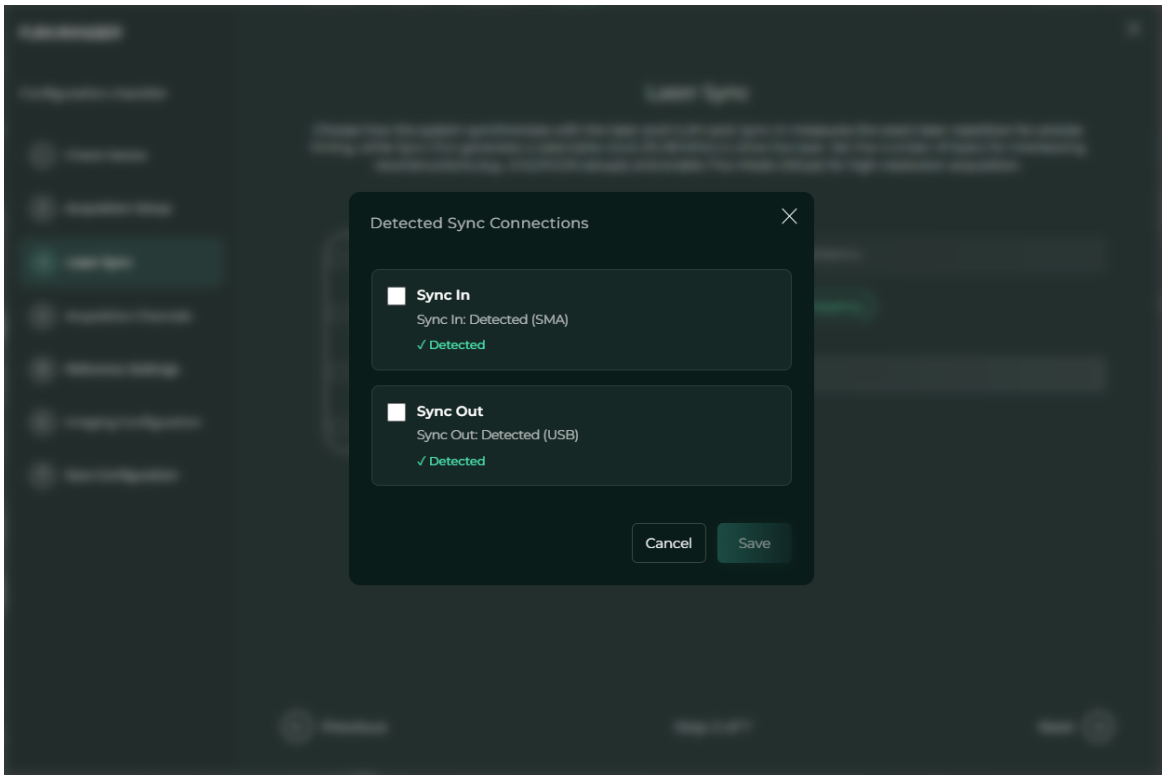
Detect connected synchronization signals by pressing Detect Sync Connections . If an external trigger signal is connected, the software can measure the frequency by clicking Measure Laser Frequency .



Select the synchronization mode to be used for timing information:

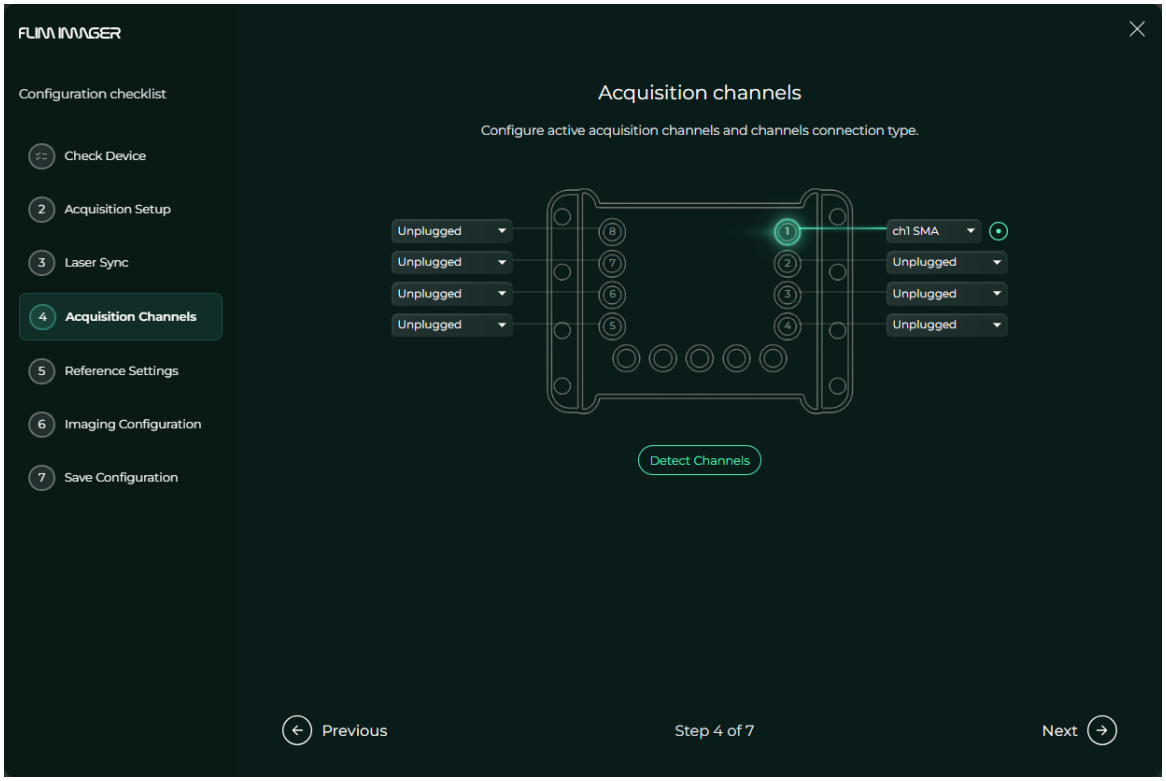
Sync In The software uses the trigger signal connected to the *sync in* port.

Sync Out The software uses the internal clock and sends a trigger signal via the *sync out* port.



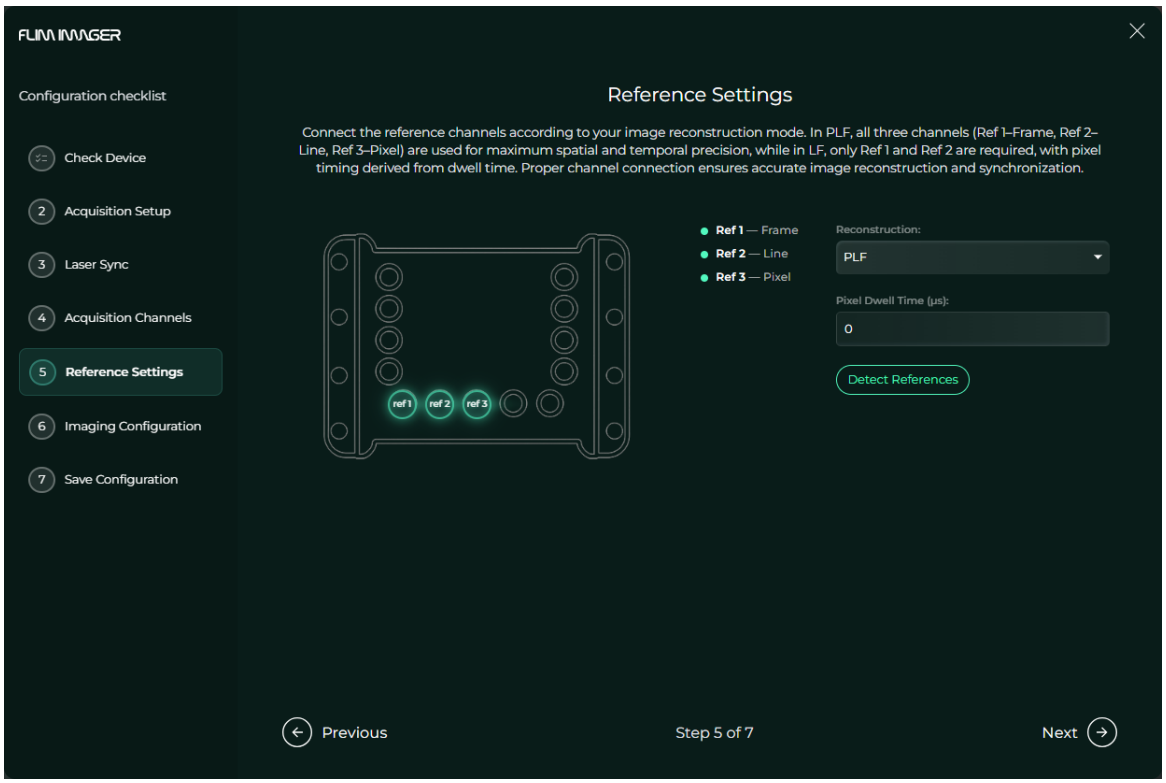
4. Choose Acquisition Channels

Press the button Detect Channels to detect connected signals automatically or choose which channels to use.



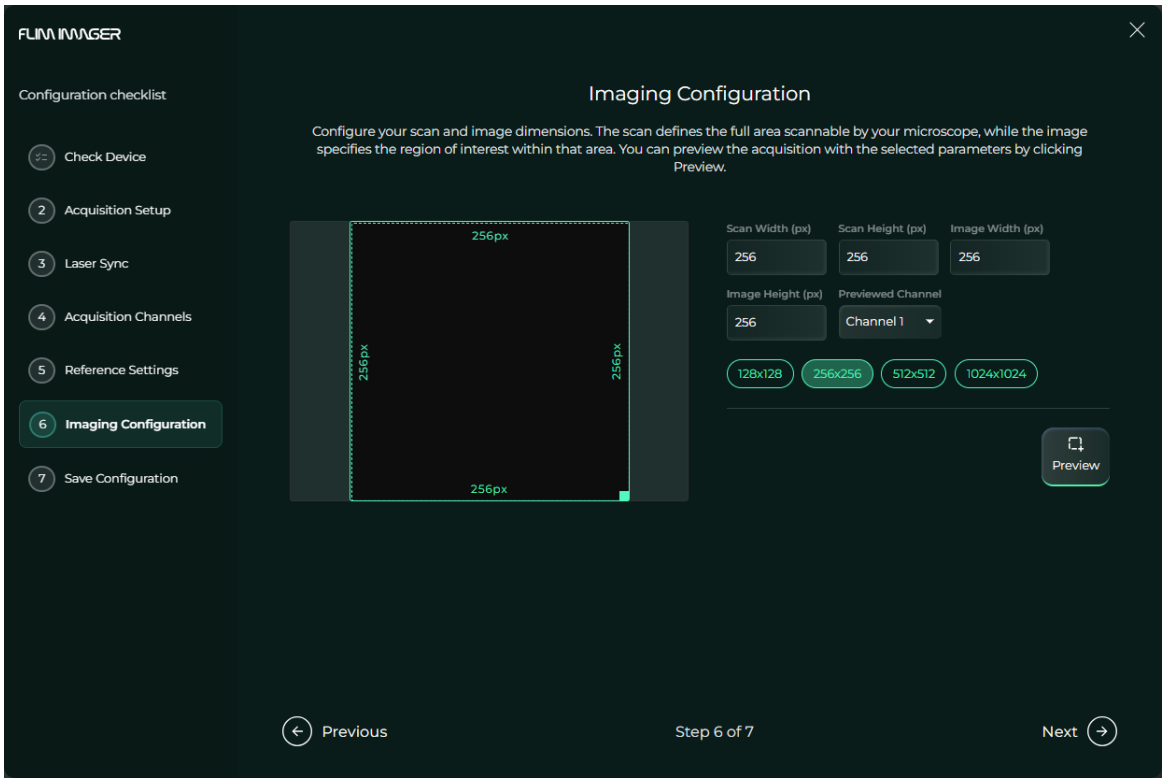
5. Define Reference Settings

Press the button Detect References to detect connected reference signals from the scanning hardware automatically or define the *reconstruction method* and the *pixel dwell time*. Refer to Section 3.6 for further information.



6. Configure Image Dimensions

Insert the scanning dimensions of your microscope setup. Users can reduce the area for recorded during acquisition by selecting a smaller frame using the mouse of by inserting specific values for image height and width. Selecting the Preview button shows a live image of the selected parameters.



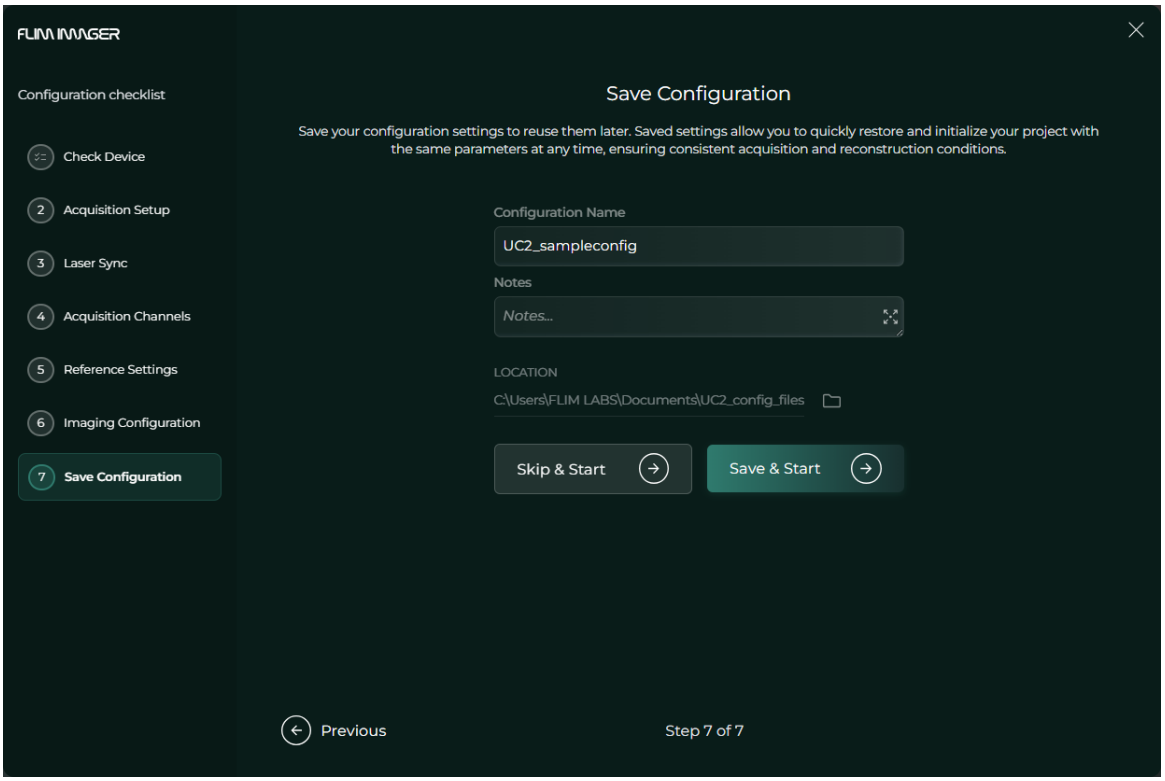
In **Live Preview** acquisition parameters can be adjusted with live feedback.



Figure 2.3: Live Preview window

7. Save Configuration

To save the configuration choose a name, the storage location, and press Save & Start . To continue without saving the settings press Skip & Start .



2.5 GUI Overview

Figure 2.4 shows the graphical user interface (GUI) of the software. The GUI provides real-time visualization of photon intensity and timing and access to all acquisition parameters and controls. Key elements of the GUI include:

1. Selection between *Scouting* mode for quick image analysis and *FLIM* data analysis mode with various extra features (see Section 2.6 and Chapter 4 for more details).
2. Acquisition information and control buttons:
   - Start for starting and stopping acquisition
   - Reset for clearing cached data and refreshing the display
   - Export for data export and time tagger activation
   - W opens a separate miniature window for convenient start/stop of acquisition (see Figure 3.5).
3. Selection of acquisition parameters and image reconstruction method (see Section 3.6).
4. The Preview button activates a live view of the image and decay profile.
5. Several configuration panels can be expanded by clicking the down arrow next to each title.
   - CONFIGURATION (see Figure 3.8)
   - EXPORT DATA (see Figure 3.6)
   - HARDWARE SETUP (see Figure 3.9)
   - CHANNELS (see Figure 3.1)
   - LASER SYNC (see Figure 3.13)
   - SCAN AREA (see Figure 3.14)
   - INTENSITY TRACING (see Figure 3.4)
6. Data display of the selected channels.

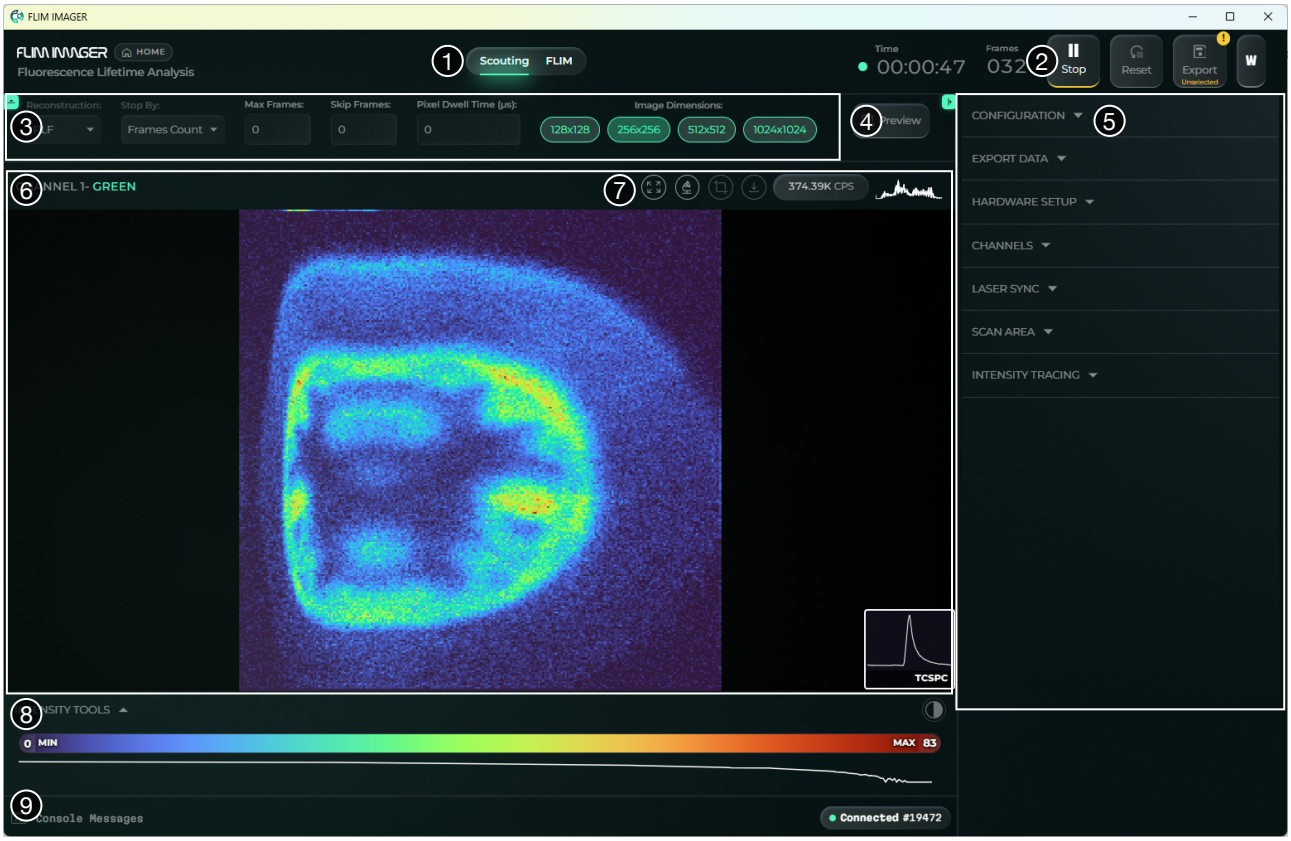


Figure 2.4: Graphical user interface of the software.

7. Display modification options. Refer to Section 3.2 for more information.
8. Graph scaling and colormap tools (see Section 3.2).
9. Console messages are. Pull-up to see messages. Refer to 5.1 for further details.

By clicking the small arrow icons , , the user can collapse panels to clean up the screen for focusing on specific areas of the interface.
The parameters and data graphs shown in the window depend on the selected data analysis mode.

2.6 Scouting & FLIM Modes

The FLIM IMAGER software provides several analysis modes for visualizing and interpreting fluorescence lifetime data. Each mode presents the acquired data in a different way, depending on the analysis objective. The modes are accessed by clicking one of the two tabs on the top of the screen:

Scouting | FLIM

Users may quickly switch between mode windows to examine the data and adjust specific settings.

Scouting Mode

Scouting Mode is preselected upon start of the software as shown in Figure 2.4. This mode is intended for quick image inspection and adjustment. The image is generated by mapping the intensity of each pixel to a color or grayscale. For this purpose, individual frames are displayed instead of accumulating subsequent frames as in regular acquisition. Additionally, a Preview button allows opening a preview window for tuning image parameters such as scan size.
For more info on display features and adjustment refer to Section 3.2.

FLIM Modes

Selecting *FLIM Mode* adds four different options as tabs shown below:

InstaFLIM | TCSPC | Calibration | Phasors .

Each mode provides access to different analysis workflows for fluorescence lifetime analysis.

InstaFLIM

In InstaFLIM mode, the reconstructed image simultaneously displays the average fluorescence lifetime and intensity information for each pixel.

- **Lifetime** is mapped to a color scale.
- **Intensity** is represented by the brightness of each pixel.

This mode provides an intuitive visualization of lifetime variations across the sample while preserving structural information through intensity contrast.

TCSPC

In Time-Correlated Single Photon Counting (TCSPC) mode, the temporal distribution of detected photons is displayed as a histogram of photon counts versus time delay.
The mode includes:

- an **intensity image** of the sample, and
- a **decay histogram** used for quantitative lifetime analysis.

This representation allows detailed inspection of fluorescence decay behavior.

Calibration

Calibration mode provides the same data visualization as TCSPC mode but includes additional controls for entering calibration parameters and acquiring reference measurements.
This mode is used to generate calibration data required for accurate lifetime or phasor-based analysis.

Phasors

Phasor mode enables graphical lifetime analysis in the phasor domain.
The display consists of:

- an **intensity image** of the sample, and
- a **phasor plot** representing the lifetime distribution of the detected signal.

This mode allows:

- analysis of lifetime components using phasor representations,
- application of calibration data,
- identification of clusters, and
- filtering of selected regions in the phasor space.

A detailed explanation of each analysis mode is provided in Chapter 4.

3 Software Operation

This chapter describes the normal workflow for operating the software after installation and initial setup.

3.1 Startup Procedure

A typical startup workflow is listed below:

1. Connect the DAQ card to the PC
2. Connect all signal and trigger cables
3. Start the software
4. Verify that the DAQ device is recognized by the software
5. Choose a saved configuration or select active acquisition channels
6. Configure acquisition settings and file storage
7. Start acquisition

3.2 Data Display

The Data Display section provides real-time visualization and interaction tools for acquired data. The type of image or graph shown depends on the analysis mode selected. A detailed explanation on the plots and features can be found in Section 2.6 and Chapter 4.

3.2.1 Channel Configuration



Figure 3.1: Channel configuration panel.

The CHANNELS panel on the right side of the screen allows users to select which input channels are acquired and displayed. A maximum of eight channels can be recorded[1]. The following options are shown:

[1] STEDYCON setups allow for only four channels with SMA input

Activate channel Enable or disable channels for acquisition by clicking the switch next to the channel number.

Add name For easier identification channel names can be added which will be shown on the data display head next to the channel number.

Toggle visibility Toggle visibility of channels in the display. A maximum of two channels can be shown simultaneously. Activated channels are recorded independently of their visibility.

Channel settings cannot be modified while an acquisition is running. To apply changes, wait until the current acquisition has finished or stop it manually.

3.2.2 Plot Interaction Features

The panel above each displayed channel provides several interaction features next to the channel number and name. During acquisition, the current count rate (CPS) is shown in counts per second, along with a small trend graph that provides a quick overview of signal stability during acquisition of the last frame. The following tools are available:

Full screen Expands the current plot to full screen mode for detailed inspection. Press *Esc* or use the same button to return to the standard view.

Magnifier Creates a square-shaped zoomed-in window that can be moved freely across the image, allowing precise inspection of local features (see Figure 3.3). Magnification values of 2x, 4x, 5x, and 10x can be selected

ROI Defines a *region of interest (ROI)* for focused analysis.
After activating the ROI tool, a selection window opens with multiple shape options:

- Rectangle
- Circle
- Ellipse
- Polygon
- Spline
- Freehand

Select a shape, draw the ROI by clicking and dragging on the image, and confirm the selection. The ROI can be repositioned by clicking and dragging it (cursor changes to a hand icon). Press *Enter* to confirm the final position.
Once applied, the display updates to show only the selected region. To return to the full image, click the ROI button again. Additional controls available in the ROI menu:

- Rotate ROI:
- Reset rotation to default orientation:

Save Image Saves the current view to a file. Clicking the download button opens a dialog where the user can specify the file name and storage location.

ROI selection and Image saving cannot be performed during active acquisition (greyed out). These options become available only after an acquisition has been completed or manually stopped.

3.2.3 Color Maps

A panel below each image shows a color scale of the data in the image together with a simple line graph indicating their frequency distribution. In the case of *InstaFLIM* mode, an additional brightness scale is shown to visualize intensity data. The range of each scale can be adjusted by clicking and dragging the *minimum* and *maximum* values shown. Hovering over the scale reveals quick-access icons for further adjustment of the image:

- Clicking the button opens an extensive list of predefined color maps for customized visualization (see Figure 3.2).
- Selecting inverts the color map for alternative contrast representation.

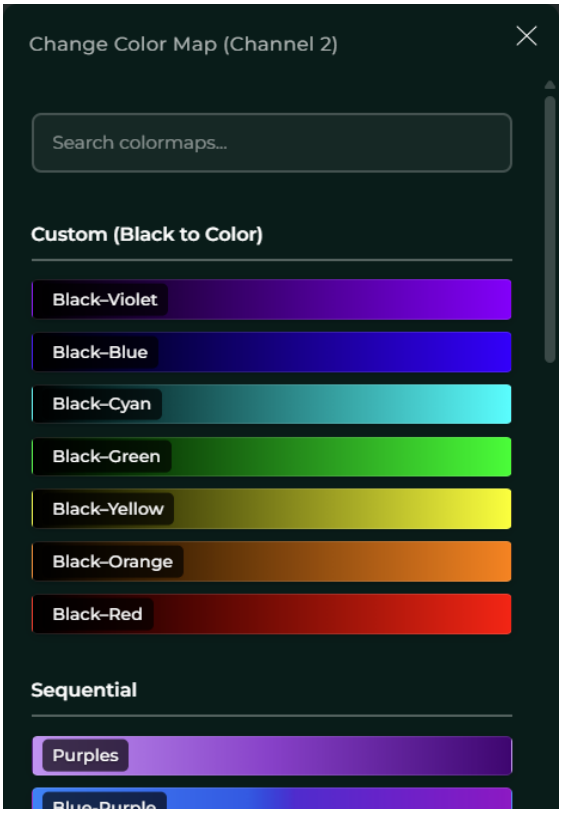


Figure 3.2: Various color maps are available for custom data visualization.

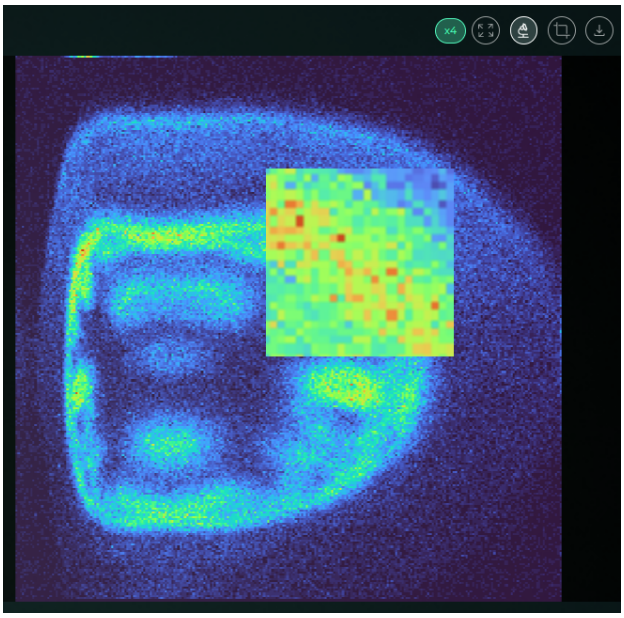


Figure 3.3: Magnifier tool for inspecting image details.

3.2.4 Intensity Tracing

Expanding the Intensity Tracing panel on the right side of the GUI allows the user to add or remove intensity features:

Show average CPS Toggle visibility of counts-per-second number above the data.

Show kcps peaks Enable or disable indicators for high count rate peaks.

Show intensity traces Toggle visibility of itensity traces above the data..

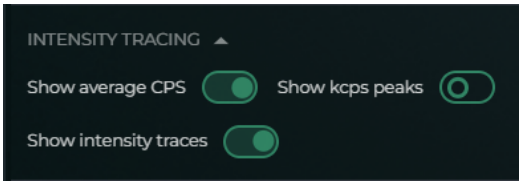


Figure 3.4: Intensity configuration

3.3 Data Acquisition

Data acquisition can be controlled using the Start button on the top right. The same button can be used to stop the acquisition as it converts to show Stop . When Timelapse is enabled, Start starts the acquisition sequence configured in the Timelapse panel.

For simple acquisition control, a separate mini-window can be opened by pressing the W button. This window can be placed conveniently to any position on your screen, for instance next to your microscope control software.

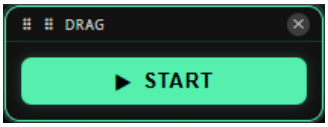


Figure 3.5: Mini-control window for convenient acquisition control.

3.3.1 Acquisition Parameters

The acquisition parameters can be found at the top of the GUI.

Duration Setting

Clicking the Stop By: drop-down menu, the user can choose between two options to define the duration of data acquisition:

Frames Count	Define the number of frames in the field Max Frames: to record after starting the acquisition.
Photon Count	Define the number of photons to record during acquisition.

Entering the value Max Frames: 0 in *Frames Count* mode will activate continuous acquisition until the user manually stops acquisition (‘*Free-running mode*’).

Skip Frames

When inserting a value for Skip Frames: , the software will skip the specified number of frames before starting data acquisition. This allows to discard any initial noisy or unstable frames, ensuring that only clean, relevant data is acquired for analysis.

Timelapse

The Timelapse ▾ panel in the right-hand sidebar allows several acquisitions to run sequentially without pressing Start for each measurement. For each acquisition in the sequence, select a stopping criterion: a specified number of frames or a specified number of photons. When Timelapse is enabled, pressing Start runs the complete sequence rather than a single acquisition. The sequence can be saved through data export and loaded again later.

3.4 Data Export

The Data Export functionality allows users to save acquired data for further analysis. Experiment parameters, notes, and tags can be included as metadata and stored alongside the exported data file.

Export settings can be configured through the panel EXPORT DATA ▾ as shown in Figure 3.6 or by clicking the Export button on the top right of the GUI.

Users can:

- Define a *file name* for the exported data
- Specify the *storage location*
- Add notes or tags for documentation
- Activate or deactivate data export using the slider .

A yellow exclamation mark and the word ‘Unselected’ are shown on the Export button to warn the user when data export is not activated. When export is enabled, starting an acquisition will open a confirmation dialog. This dialog allows the user to review and confirm the selected export settings before the acquisition begins.

Explanation on how to process exported data using the provided *Python* and *MATLAB* scripts is provided in Section 5.2. A description of the file formats is provided in Chapter 8.

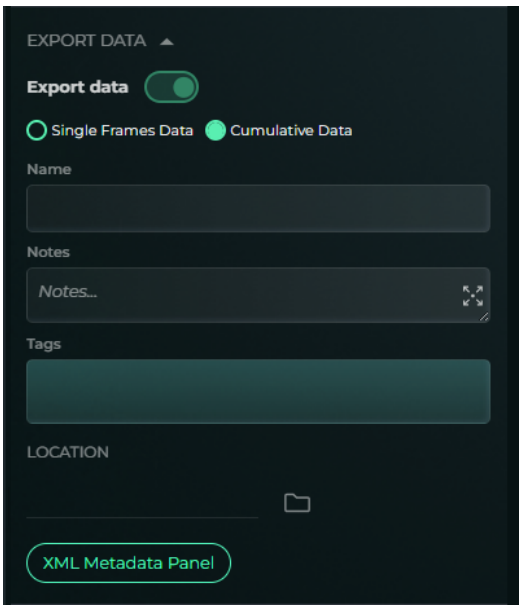


Figure 3.6: Data export configuration panel.

After configuring the data export settings make sure to activate data export by enabling the *Export data* .

Export Options

In scouting mode, only single frame data is shown and exported. In other analysis modes, users can choose between two data export options:

Single Frame Data	Exports individual frames as separate datasets, preserving time-resolved information for each acquisition step.
Cumulative Data	Exports the accumulated data over the entire acquisition, providing a single dataset representing the integrated measurement.

Metadata

Additional experiment-specific information can be entered such as parameters, comments, and other contextual data relevant to the measurement. This data is stored in a separate file .XML file called `{name}_{timestamp}_metadata.xml` .

Including metadata in your data files can help ensure traceability and simplify post-processing and data interpretation.

3.5 Configuration Options

3.5.1 Configuration Selection

After completing the initial setup using the *Configuration Wizard* users can access the configurations expanding the CONFIGURATION panel on the right side of the GUI. This panel shown in Figure 3.8 displays the current active configuration and allows to start the *Configuration Wizard* again or select a different configuration file.

3.5.2 Hardware Setup

Selecting HARDWARE SETUP opens a panel that displays the active FLIM Data Acquisition Card and options for activating multiple hardware diagnostics functions:

Detect Cards	The software will scan for any FLIM DAQ cards connected to the PC.

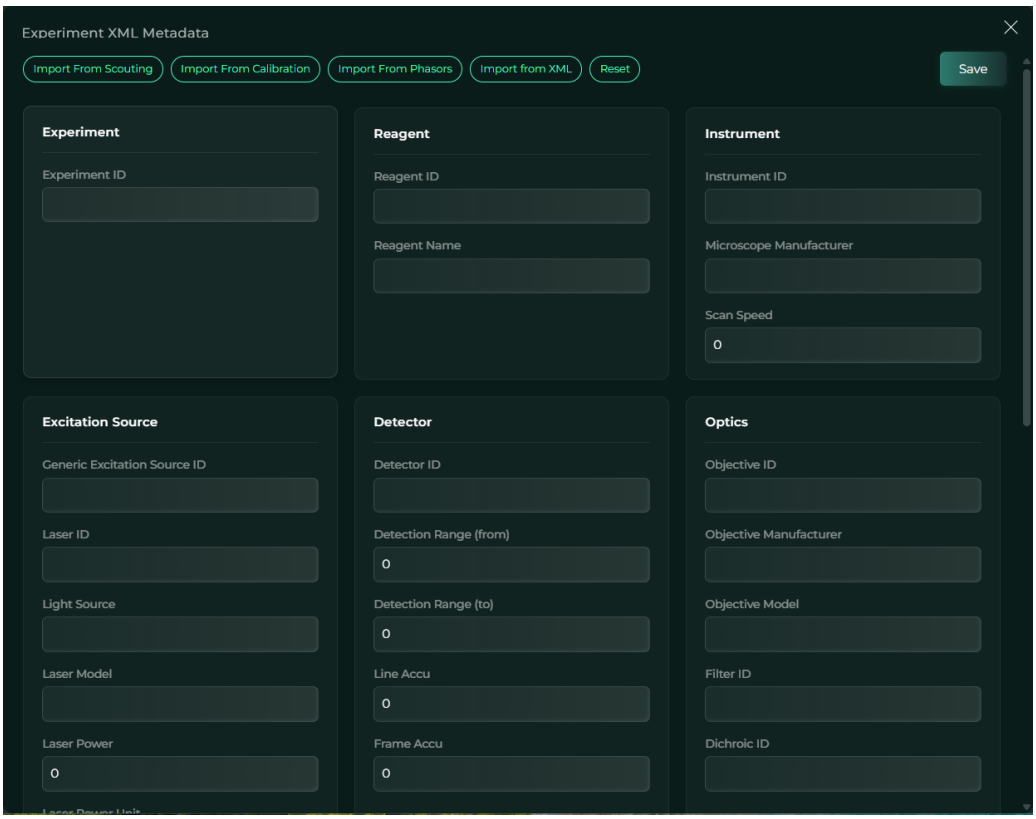


Figure 3.7: Interface for adding experiment-specific metadata stored as XML in the exported data file.

Detect Connections The software will check all ports for any connections for synchronization, reference, and data input signal and open a window as shown in Figure 3.10.

Scanning Diagnostic This diagnostic tool measures various scanning signal properties (see Figure 3.11).

Imaging Diagnostic The Imaging diagnostic shown in Figure 3.12 will open a window displaying the hardware connections for image generation and synchronization.

3.5.3 Laser Synchronization

Selecting the panel LASER SYNC opens a window for selecting one of the laser synchronization options. Clicking Sync In opens another window for measuring a trigger signal connected to the *Sync IN* port of the DAQ.

3.5.4 Scan Area

The scan area and image size are configured in the panel SCAN AREA (see Figure 3.14). The parameters *Image Width* and *Image Height* define the region of the available scan area that will be acquired. This allows users to restrict acquisition to a smaller region of interest instead of the full scan field. The image area is represented by a dark area enclosed by a dotted line in the window. Position and size of the image area can also be adjusted by dragging and resizing the frame using the mouse. Opening the *Preview Mode* (see Figure 2.3) allows for direct feedback of the selected area.

Laser galvanometer scanning systems typically cause image distortion at the edges of the scanning area. Therefore, it can be useful to choose a larger scanning area of the sample and then discard the distorted areas by selecting a smaller image size.

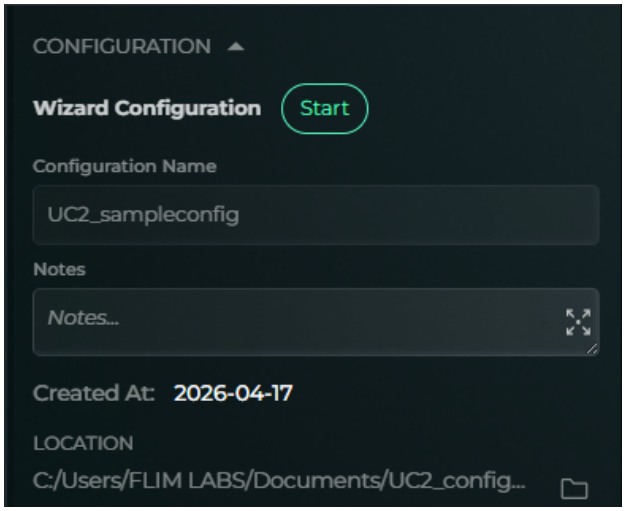


Figure 3.8: The Configuration panel displays the active configuration and allows the user to start the *Configuration Wizard*.

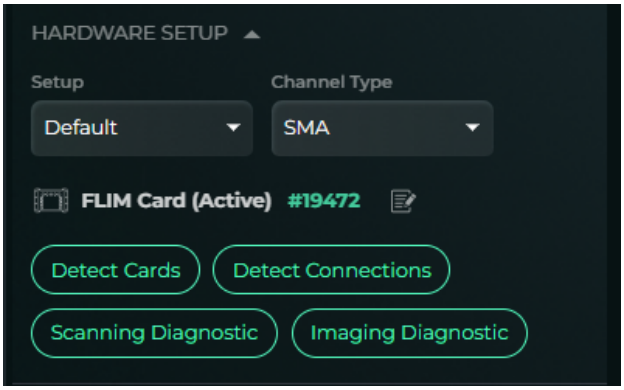


Figure 3.9: Hardware setup

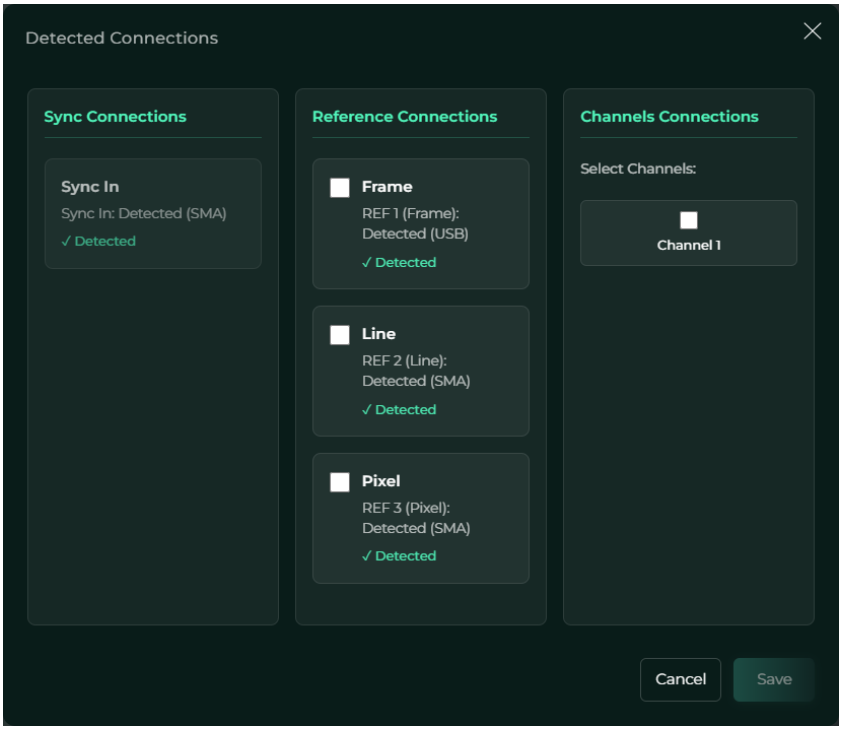


Figure 3.10: Detected Connections

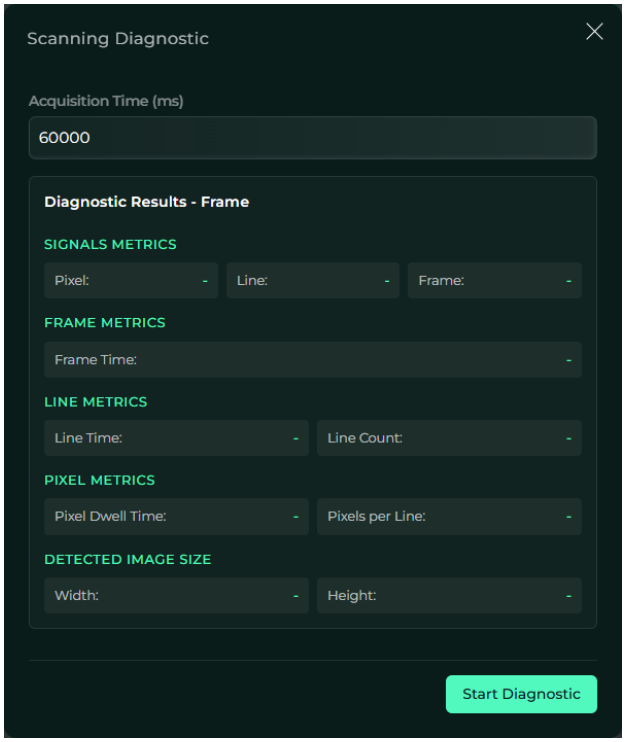


Figure 3.11: Scanning diagnostic window.

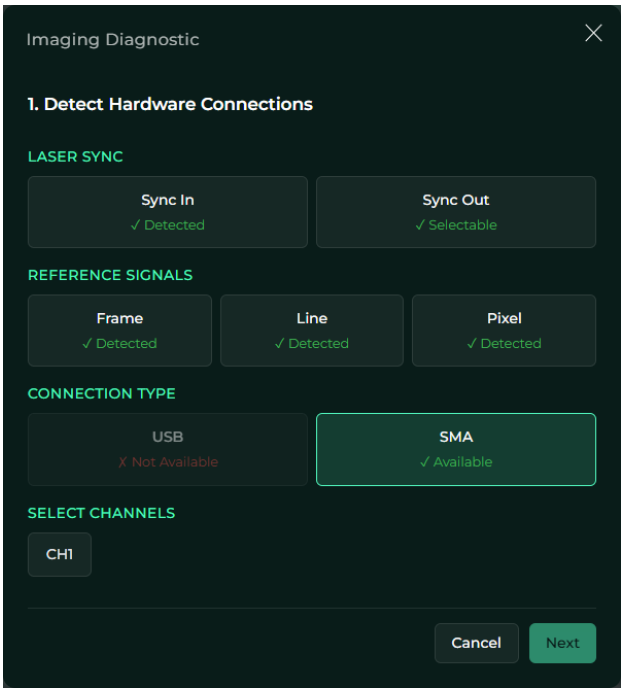


Figure 3.12: Imaging diagnostic window.



Figure 3.13: Laser synchronization settings

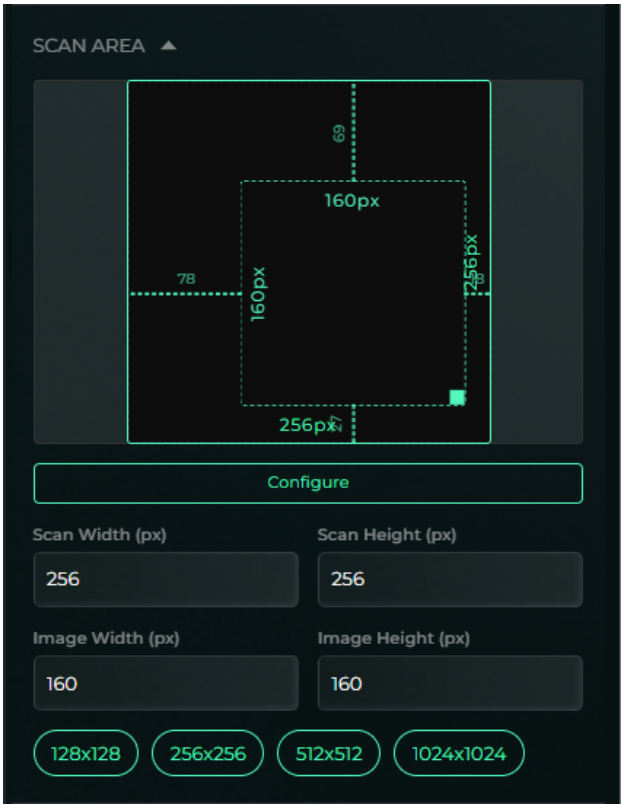


Figure 3.14: Adjust scan area

3.6 Image Reconstruction

The FLIM IMAGER software reconstructs images from the detected fluorescence signal using synchronized scanning information. Detected photon events are assigned to spatial positions based on pixel, line, and frame timing signals provided by the scanning system. Alternatively, simplified reconstruction methods can be used, which rely on user-defined parameters instead of full synchronization signals.

Reconstruction Principle

Image reconstruction relies on the combination of:

- detected fluorescence signal,
- synchronization signals from the scanning system,
- user-defined spatial parameters such as image size and scan range.

Depending on the selected reconstruction mode, the software uses the available synchronization inputs (pixel, line, and/or frame clocks) to assign detected events to their corresponding image coordinates. Figure 3.15 shows a schematic explaining the three clock signals for pixel, line, and frame.

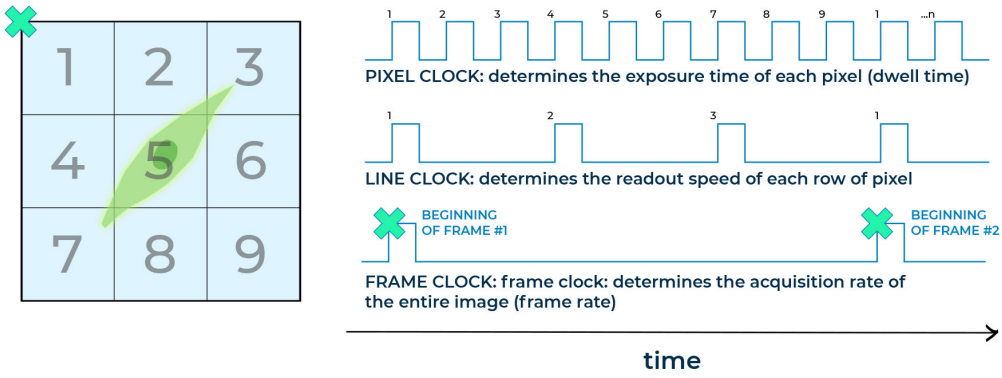


Figure 3.15: Illustration for spatial and temporal arrangement of pixel, line, and frame timing signals that allow for unambiguous assignment of fluorescence data to a specific sample location.

Reconstruction Modes

The user can choose from three reconstruction methods:

PLF (Pixel–Line–Frame)
: Standard reconstruction mode. Uses full synchronization signals (pixel, line, and frame) to achieve accurate spatial mapping. This mode is recommended for most applications.

LF (Line–Frame)
: Simplified reconstruction mode using line and frame synchronization. Pixel timing is reconstructed based on a fixed dwell time.

F (Frame only)
: Minimal reconstruction mode using only frame synchronization. Pixel and line timing are inferred from the specified dwell time.

Simplified reconstruction modes (LF or F) are used when pixel clock synchronization is not available or cannot be reliably acquired. They are also suitable for preliminary measurements or system diagnostics where approximate spatial reconstruction is sufficient.

Pixel Dwell Time

When using simplified reconstruction modes (**LF** or **F**), the pixel dwell time must be specified manually. This parameter defines the time spent on each pixel during scanning and is required to correctly reconstruct the image.

The pixel dwell time can be determined directly within the software using the scanning diagnostic function available in the *Hardware Setup* panel. See Section 3.5.2 for more details.

For accurate image reconstruction, ensure that the selected reconstruction mode matches the available synchronization signals of your scanning system. Incorrect configuration or inaccurate pixel dwell time may result in distorted or misaligned images.

Scanning Hardware

At the hardware level, scanning can be achieved either by moving the laser beam across the sample with scanning mirrors and a focusing lens or by moving the sample with a motorized stage while the laser beam remains fixed. Figure 3.16 illustrates both methods. In both cases, the sample is scanned pixel by pixel, and the emitted fluorescence is detected and sent together with the spatial synchronization signals to the data acquisition system. This synchronization is essential because it allows the software to associate each measured signal with its correct position in the reconstructed image.

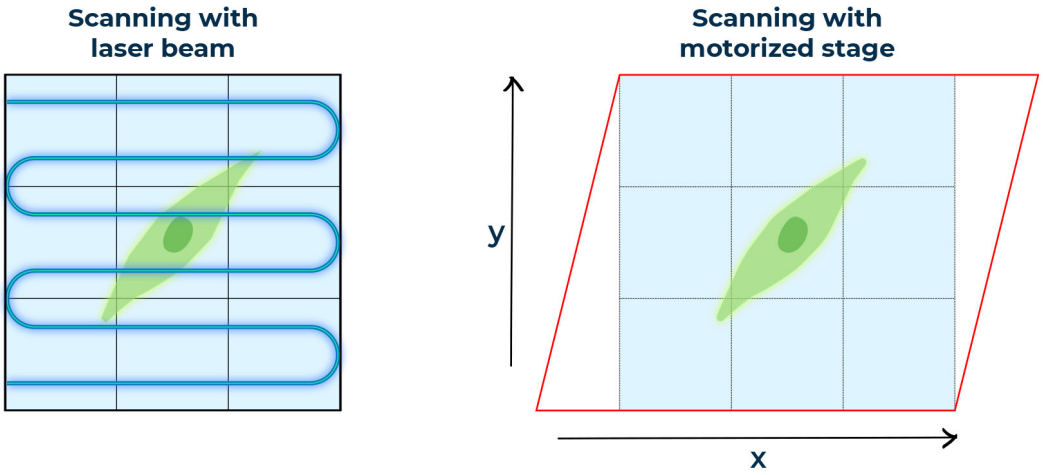


Figure 3.16: Schematic representation of hardware solutions of pixel-by-pixel scanning for image reconstruction.

4 FLIM Analysis Modes

This chapter provides a detailed description of the available FLIM analysis modes and their specific functionality.

4.1 InstaFLIM

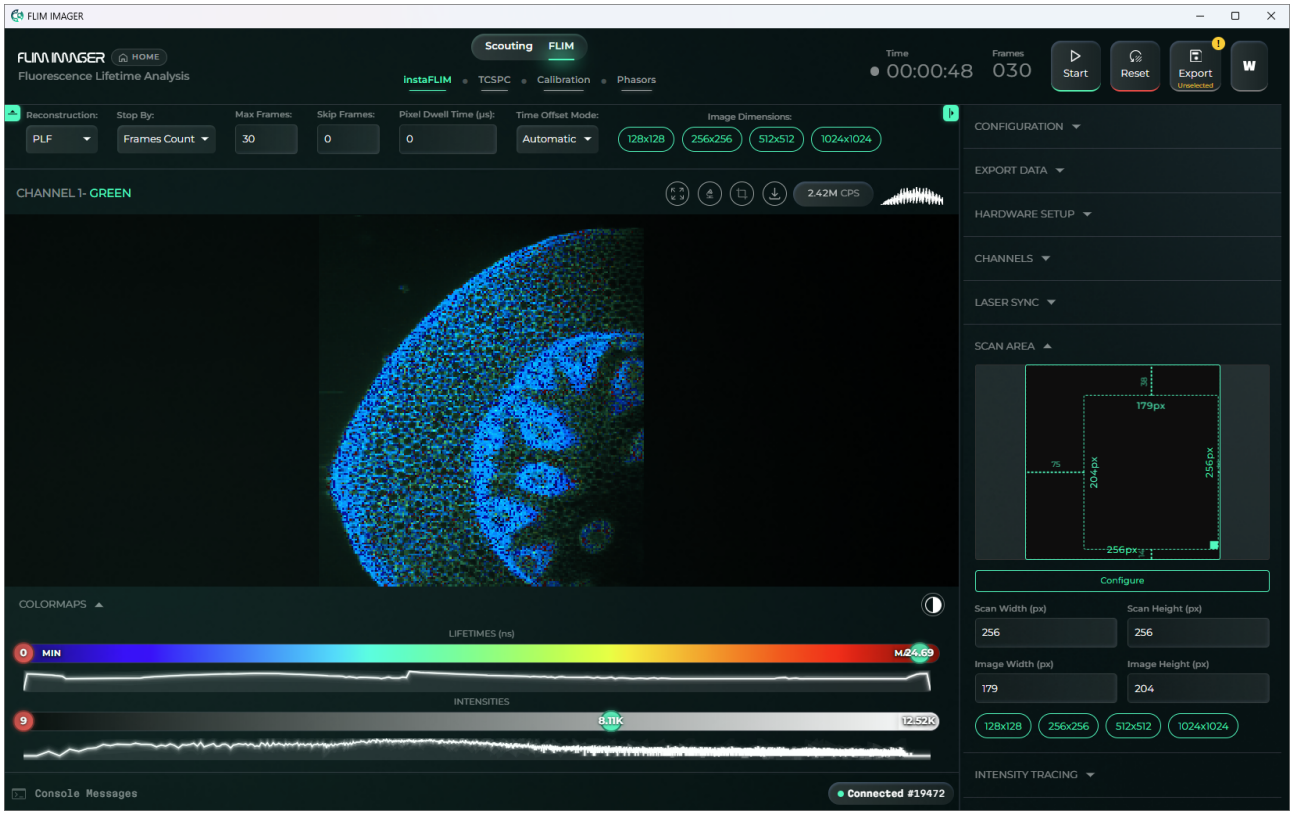


Figure 4.1: InstaFLIM mode.

In InstaFLIM mode, fluorescence lifetime and intensity information are combined into a single image representation.
For each pixel:

- the **fluorescence lifetime** is mapped to a color scale,
- the **intensity** is represented by the pixel brightness.

This mode provides a direct and intuitive visualization of spatial lifetime variations while preserving structural information through intensity contrast. It is particularly suited for rapid inspection and qualitative analysis of samples.

Time Offset Mode

For an accurate lifetime calculation, the decay profile as recorded by the DAQ needs to be shifted in time. By selecting the drop-down menu Time Offset Mode: , the user can select one of two modes:

Automatic	The software estimates the a value for the time shift.
Manual	An additional field Time Offset (ns) is shown allowing the user to enter a specific value.

4.2 TCSPC

In Time-Correlated Single Photon Counting (TCSPC) mode, the temporal fluorescence decay is displayed as a histogram of photon counts versus time delay, alongside the intensity image.



Figure 4.2: TCSPC mode.

Display Options

- The *signal-to-background ratio (SBR)* is displayed in the upper right corner of the histogram. The SBR quantifies signal quality by comparing the peak fluorescence intensity with the background noise level. SBR is measured in decibels (dB) and helps to assess the reliability and contrast of the data.
- The histogram can be displayed on *linear* or *logarithmic* scale by selecting the Linear | Log button below the histogram.
- The decay curve can be shifted in time by:
  - adjusting the slider below the histogram, or
  - entering a specific value in nanoseconds in the field next to the slider.

Graph Interaction

Hovering over the TCSPC plot reveals additional interaction tools:

- save image,
- zoom,
- center,
- reset view
- zoom specific area
- panning

Region-Based Analysis

Selecting individual pixels or regions of interest (ROI) updates the TCSPC histogram to display the corresponding decay. ROIs can be added clicking on the button above the image. A detailed description of ROI selection can be found in Section 3.2.



Figure 4.3: Calibration mode.

4.3 Calibration

Calibration mode is used to acquire reference data for accurate lifetime and phasor analysis. A reference sample with a known mono-exponential decay time τ is required for this process. FLIM LABS provides dedicated calibration samples in the **FLIM-Phasors Calibration Kit**. For more information please check the website under https://www.flimlabs.com/products/phasors-calibration-kit.

Calibration Procedure

1. Insert the reference sample in your imaging setup.
2. Enter the corresponding decay constant in the field Tau (ns) in the parameter panel above the plot area.
3. Select appropriate acquisition parameters.
4. Start the acquisition using the Start button.



The imaging and illumination parameters used during calibration should match those of the actual measurement to avoid calibration mismatch.
If the detected signal exceeds the saturation threshold of the detector, reduce the intensity using neutral density (ND) filters or by defocusing the excitation beam.

When calibration is complete, the software saves a separate reference file in addition to the image. Its filename ends with `imaging_calibration.json`, and it is stored in `/.flim-labs/data/`.

This file contains the reference sample lifetime, the laser frequency, and the correction values used for phasor analysis. Keep it with the corresponding measurement files so that the calibration can be reused later.

The acquired calibration data can be applied in the *Phasors* mode panel.

4.4 Phasors

Phasors mode enables graphical analysis of fluorescence lifetime data in the phasor domain. It provides an intuitive way to interpret lifetime distributions without requiring explicit curve fitting.



Figure 4.4: Phasor mode.

It is particularly useful for identifying mixtures of fluorophores or environmental changes within a sample.
The interface displays:

1. an intensity image of the sample, and
2. a phasor plot representing the lifetime characteristics of the detected signal.

Calibration

Previously recorded calibration data can be loaded using the CALIBRATION panel above the phasor plot:

- Select Use Calibration Tab Reference to apply the calibration acquired in the current session.
- Load a previously saved calibration using the Load button and select the file whose name ends with `imaging_calibration.json`.

Phasor Plot

The phasor plot represents the fluorescence decay of each pixel as a point in a two-dimensional coordinate system defined by the components G and S. Each point corresponds to a specific lifetime behavior:

- Mono-exponential decays lie on the universal semicircle.
- Multi-exponential decays appear inside the semicircle as linear combinations of lifetime components.

This representation allows direct visual identification of different fluorescent species or environments within the sample.

Display Options and Filters

The signal-to-background ratio (SBR) is indicated above the Phasor plot. Three additional display options can be selected below to improve visualization and reduce noise:

- Toggle visualization of fiducial points using the switch *Fiducial Points*

- Select Median filter for applying a spatial median filter to the data
- Select pawFLIM for lifetime-aware denoising

Filters can be applied successively by repeated activation. All applied filters can be reset using the circle button ↺.

Cluster Analysis

The PHASOR CLUSTERS ▾ panel allows grouping of data points directly in the phasor plot for spatial interpretation.
To create a new cluster click the Cluster + button.

1. Choose one of the following geometric shapes:
   - circle,
   - ellipse,
   - rectangle,
   - freehand.
2. Click once in the Phasor plot to define the center or starting point of the cluster position.
3. Click again to define the radius or continue drawing the geometric shape.
4. Confirm the shape by pressing enter.

Each cluster added will be listed as new row in a table shown below the Phasor plot:

Color	Show	Alpha	Radius	Thick	G	S	τ_φ (ns)	τ_n (ns)	τ_m (ns)	m	φ
	☑	☐	75.50	1	0.316	0.457	5.47	6.08	6.34		

Selected clusters are mapped back to the image, highlighting the corresponding regions in the sample with a unique color. Values in framed boxes can be directly modified by clicking and modifying values in the table.
All clusters can be removed using the Reset All ↺ button.

Additional Phasor Segmentation Tools

In addition to PHASOR CLUSTERS ▾, the Phasors screen provides the Colormaps ▾ and 2-Components ▾ tabs. The 2-Components ▾ tab provides an A–B segment for two-component analysis.

5 Advanced Operation

5.1 Console Messages

The software includes a log console at the bottom of the application window for displaying diagnostic and status messages. By default, this console may not be visible because it is collapsed at the lower edge of the interface. To display the messages, move the mouse to the lower border of the application window and drag the console area upward.

The log console keeps a history of up to 500 messages. Each entry is timestamped and classified by message type, including:

- Info messages (white),
- Warning messages (yellow),
- Success messages (green),
- Error messages (red).

These messages provide feedback about system status, configuration changes, data acquisition, export actions, and possible errors. The console can therefore be used to monitor software behavior and to assist in troubleshooting during acquisition and analysis.

5.2 Data Processing Scripts

When exporting data, FLIM IMAGER automatically generates scripts for *Python* and *MATLAB*. These scripts are stored together with the acquisition file and enable efficient processing and visualization of the exported `.json` data files.

The scripts provide a reference workflow to:

- parse binary data into structured arrays,
- reconstruct images from photon timing data,
- visualize intensity images and lifetime-related information,
- adapt reconstruction and analysis parameters for custom workflows.

In contrast to simple intensity tracing, the exported data contains spatial and temporal information. Therefore, correct visualization requires applying the appropriate reconstruction method and acquisition parameters, which are included in the metadata and exposed in the scripts.

Script Configuration

The generated Python and MATLAB scripts contain an initial configuration section with variables that can be adjusted before running the script. These variables define which exported file is processed and how the data is visualized.

Depending on the script type, the following configuration variables may be available:

file_path	Defines the imaging `.JSON` file to be analyzed. This value can be changed to process a different exported imaging file.
imaging_file_path	Defines the imaging `.JSON` file used by the single-frame phasors script. This variable is only present in the single-frame phasors script.
phasors_file_path	Defines the phasors `.JSON` file to be analyzed. This variable is only present in phasors scripts.
channel	Selects which active acquisition channel is visualized. By default, the first active channel is selected, but the value can be changed to any other active channel.
colormap	Defines the colormap used for image reconstruction. The default colormap is `hot`.
frame	Selects the frame number to analyze. This variable is only present in the single-frame reconstruction script.

harmonic Selects the phasor harmonic to analyze. This variable is only present in phasors scripts.



Only the variables relevant to the selected export type are included in the generated script. For example, phasors scripts include phasor-specific file paths and harmonic selection, while single-frame scripts include frame selection.

Python

To visualize and process the exported data using Python:

1. Open a terminal and navigate to the export directory:
   ```
   cd YOUR_DOWNLOADED_DATA_ROOT_FOLDER
   ```
2. Create a virtual environment:
   ```
   python -m venv venv
   ```
3. Activate the virtual environment:
   ```
   venv\Scripts\activate
   ```
4. Install the required dependencies:
   ```
   pip install -r requirements.txt
   ```
5. Run the generated script:
   ```
   python YOUR_SCRIPT_NAME.py
   ```

The Python script reconstructs the image data from the raw photon timing information and displays the resulting intensity image. Depending on the configuration, additional outputs such as decay data or intermediate reconstruction results may be available.

MATLAB

To visualize and process the exported data using MATLAB:

1. Open the MATLAB command window.
2. Navigate to the export directory:
   ```
   cd YOUR_DOWNLOADED_DATA_ROOT_FOLDER
   ```
3. Run the generated script:
   ```
   YOUR_SCRIPT_NAME
   ```

The MATLAB script performs equivalent reconstruction and visualization steps, allowing users to analyze the data within the MATLAB environment and integrate it into existing workflows.

6 Troubleshooting

6.1 Common Issues

Table 6.1: Common issues, causes, and solutions

Issue	Cause	Solution
No acquisition card detected	The acquisition card is not connected, not powered correctly, or was not detected during software startup	Check all USB and signal connections, verify the device power supply, use the Detect Card button, and restart the software if necessary.
Distorted image near image edges	Scanning artifacts caused by galvo scanner nonlinearity or overscan region	Reduce the image area or adjust the scanning region to avoid distorted edge areas.
No trigger signal detected	Microscope scanning has not started or trigger signal is missing	Verify that the microscope scan is active and that the trigger cable is connected correctly.
No data acquisition	Incorrect synchronization or trigger configuration	Verify that the selected SYNC IN / SYNC OUT configuration matches the physical cable connections and microscope trigger setup.
Low photon count or dark image	Insufficient fluorescence signal or detector settings not optimized	Increase excitation intensity within safe limits, verify detector alignment, and check acquisition settings.
Detector saturation	Photon count rate exceeds detector or acquisition limits	Reduce excitation intensity, use ND filters, or decrease detector gain.
Image updates slowly	Large image size, high dwell time, or insufficient computer performance	Reduce image resolution, lower dwell time, or close unnecessary background applications.
Phasor plot appears noisy	Insufficient photon statistics	Increase acquisition time or excitation intensity to improve photon count statistics.
Exported files missing	Export path invalid or export disabled	Verify the selected export directory and confirm that data export is enabled before acquisition.

7 Safety, License and Support

7.1 Safety and Operational Limitations

The software does not perform hardware safety checks. Users are responsible for correct configuration and operation.

For detailed safety instructions, refer to the user manual of the FLIM Data Acquisition Card: https://www.flimlabs.com/products/FLIM-DAQ

7.2 License

This software is distributed under the MIT License. It permits free use, modification, and redistribution, including for commercial applications, provided that the original copyright notice and license disclaimer are included.

7.3 Contact and Support

For additional information, visit: https://www.flimlabs.com/products/FLIM-Imager/

For technical support, contact FLIM LABS at: `info@flimlabs.com`

8 Technical Reference

8.1 File Format

The FLIM IMAGER software exports analyzed imaging data in JSON format. The exported files contain both metadata and measurement data, allowing the acquisition to be reconstructed and visualized outside the software using the provided Python or MATLAB scripts.

The imaging and phasor measurement JSON files described below follow the same general structure:

1. **Header metadata**
   The header contains information required to identify the file type and interpret the exported data.
2. **Data records**
   The data section contains the exported imaging or phasor data. The exact structure depends on whether the file contains imaging data or phasor data, and whether the export represents a single frame or cumulative data.

Header metadata fields include the following; some are specific to the acquisition mode or setup:

- `type` : File type, such as `Frame` or `Global` .
- `file_id` : ASCII values identifying the specific file format.
- `setup` : Microscope setup, such as `Default` or `STEDYCON` .
- `abberior_multichannel_assignment_mode` : STEDYCON multi-channel reconstruction assignment mode. This field is only present in STEDYCON setup mode.
- `channels` : Array of eight booleans indicating which physical acquisition channels were enabled. Index 0 corresponds to channel 1 and index 7 to channel 8.
- `laser_period_ns` : Laser period in nanoseconds.
- `step` : Imaging experiment type, such as `scouting` , `calibration` , `tcspc` , `phasors` , or `insta-flim` .
- `reconstruction` : Image reconstruction method.
- `image_width` : Image width in pixels.
- `image_height` : Image height in pixels.

Export Schema and Additional Header Fields

The fields below belong to the JSON header. Their presence depends on the file type, acquisition mode, or reconstruction mode. The `roi_masks` field is stored separately alongside the header, as described below.

String identifying the export schema version, currently `"1.4"` . It is present in all four measurement file types: single-frame imaging (`IMF1`), cumulative imaging (`IMG1`), single-frame phasors (`IPF1`), and cumulative phasors (`IPG1`). It is not the software version; the file type is identified by `file_id` . Generated reader scripts compare the schema version with the value they expect and stop if the values do not match. The separate calibration reference file does not contain `export_version` .

Object identifying the physical input represented by the file. It contains `chIndex` , a number from 0 to 7 corresponding to an index in `channels` , and `chNumber` , the same channel numbered from 1 to 8 as displayed in the GUI. This field is present in single-frame imaging files and both phasor file types. It is absent from cumulative imaging files, which contain all acquired channels together.

UUID string generated only for Phasors acquisitions, with one identifier per channel and frame. The same identifier is written to the single-frame imaging file and its matching phasor file. Pair files using the same `pairing_id` , channel, and frame. In other acquisition modes and in cumulative files, this field is absent or `null` .

Array of eight numbers in nanoseconds, written only for TCSPC acquisitions. Index 0 corresponds to channel 1 and index 7 to channel 8. Each value is the time shift selected in the GUI, converted from bins to nanoseconds. Exported counts have not been shifted. Apply the shift when reading the file, as the generated scripts do; a positive value moves the curve toward later times. In other acquisition modes the field is absent or `null`; it is not present in phasor files.

Number in microseconds, or `null`. In Line-Frame and Frame reconstruction, this is the dwell time entered by the user, not a measured value. In Pixel-Line-Frame reconstruction, the GUI input is disabled and the measured dwell time is used only for display; the exported field is absent or `null`.

Number in nanoseconds, or `null`, describing the delay from the start of the laser period to the start of the fluorescence rise. The offset is non-negative and moves time zero forward by that amount. InstaFLIM calculates the displayed lifetime as the bin time minus this offset, wrapped within the laser period. Exported histograms have not been corrected; apply the offset only when recalculating the lifetime. This field is written for InstaFLIM acquisitions in single-frame and cumulative imaging files. An offset of zero is not stored. The field is not present in phasor files.

ROI Masks and Pixel Ordering

The `roi_masks` field is stored alongside the header, not inside it. When masks are required, it is an array with one entry per exported channel. Each entry is either `null` or a flat array of integers with one value per pixel in that channel's crop. Its length is the crop width multiplied by the crop height, as specified in `channels_dimensions`.

Pixels are ordered row by row, from left to right and then from top to bottom, beginning at the top-left corner of the rectangle enclosing the drawn ROI. Mask values have the following meanings:

`export_ve` [illegible] `offsets_ns` inside the drawn ROI;

- -1: the pixel is inside the enclosing rectangle but outside the drawn shape.

When no ROI is defined, or the ROI is an axis-aligned rectangle, no mask is needed and `roi_masks` is an empty array. Counts for pixels outside the drawn shape are zero. Phasor G and S coordinates are still written for these pixels; exclude them using the ROI mask rather than searching for -1 in the coordinate arrays.

The following sections describe the individual exported file types.

Single-Frame Imaging File Format

The single-frame imaging file stores decay data for one acquired image frame.

The header contains the common metadata fields listed above. In this format:

- `type` is set to `Frame`.
- `file_id` corresponds to `IMF1`.

Following the header, the `data` field contains decay values for the pixels in the frame. The data is structured as a three-level array:

1. The first level contains one subarray for the acquired data acquisition channel.
2. The second level contains one subarray for each pixel in the acquired frame.
3. The third level contains bin-count pairs for each pixel. Each pair has length 2:
   - the first value is the bin index,
   - the second value is the photon count for that bin.

Pixels are binned into 256 temporal bins. Bins with zero photon counts are omitted to reduce file size.

Cumulative Imaging File Format

The cumulative imaging file stores the summed imaging data across all acquired frames.

The header contains the common metadata fields listed above. In this format:

- `type` is set to `Global`.
- `file_id` corresponds to `IMG1`.
- `frames` stores the total number of acquired frames.

Following the header, the `data` field contains decay values for the pixels in the cumulative image. The data is structured as a three-level array:

1. The first level contains one subarray for each active acquisition channel.
2. The second level contains one subarray for each pixel in the acquired image.
3. The third level contains bin-count pairs for each pixel, where:
   - the first value is the bin index,
   - the second value is the photon count for that bin.

As in the single-frame imaging format, pixels are binned into 256 temporal bins and zero-count bins are omitted. In the cumulative imaging file, photon counts are the cumulative sum of the counts from all acquired frames.

Single-Frame Phasors File Format

The single-frame phasors file stores phasor data calculated for one acquired frame.
The header contains the common metadata fields listed above. In this format:

- `type` is set to `Frame`.
- `file_id` corresponds to `IPF1`.
- `frames` stores the number of acquired frames.
- `tau_ns` stores the reference lifetime value in nanoseconds.
- `harmonics` stores the number of phasor harmonics.

Following the header, the `data` field contains:

- `frame`: Number of the last acquired frame.
- `channel`: Acquisition channel used for the phasor calculation.
- `harmonic`: Harmonic used for the phasor calculation.
- `g_data`: Array containing the G coordinate for each pixel.
- `s_data`: Array containing the S coordinate for each pixel.

The arrays `g_data` and `s_data` contain one value per pixel and represent the phasor coordinates of the fluorescence decay.

Cumulative Phasors File Format

The cumulative phasors file stores phasor data calculated from the globally acquired image.
The header contains the common metadata fields listed above. In this format:

- `type` is set to `Global`.
- `file_id` corresponds to `IPG1`.
- `frames` stores the total number of acquired frames.
- `tau_ns` stores the reference lifetime value in nanoseconds.
- `harmonics` stores the number of phasor harmonics.

Following the header, the `phasors_data` array contains dictionaries with:

- `frame`: Number of the last acquired frame.
- `channel`: Acquisition channel used for the phasor calculation.

- `harmonic` : Harmonic used for the phasor calculation.
- `g_data` : Array containing the G coordinate for each pixel.
- `s_data` : Array containing the S coordinate for each pixel.

In addition, the `intensities_data` field contains decay values for the pixels in the acquired image. This field is structured as a three-level array:

1. The first level contains one subarray corresponding to the active acquisition channel.
2. The second level contains one subarray for each pixel in the acquired image.
3. The third level contains bin-count pairs for each pixel, where:
   - the first value is the bin index,
   - the second value is the photon count for that bin.

As with cumulative imaging data, photon counts in `intensities_data` are cumulative sums over the acquired frames.



Masked pixels are identified by -1 in the corresponding `roi_masks` entry. Phasor G and S values are still exported for these pixels. Exclude them using the mask; do not search for -1 in `g_data` or `s_data`. See the ROI mask description in this chapter.



The exported JSON file type determines which data fields are present. Imaging files contain decay data in the `data` field, while phasors files contain phasor coordinate arrays and, for cumulative phasors exports, additional intensity decay data.

Calibration Reference File

The calibration reference is saved separately from the imaging data in a file whose name ends with `imaging_calibration.json`. The file is stored in `/.flim-labs/data/` and contains the reference sample lifetime, laser frequency, and correction values used by the phasor plot.
To reuse this calibration, select the file using the Load button in the CALIBRATION panel on the Phasors screen. Archive this reference file together with the associated measurement data.